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Моделирование дивидендных гэпов с помощью случайных процессов с прыжковым элементом

Modelling Dividend Gaps with Jump Stochastic Processes

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Abstract

The aim of this paper is to propose a number of numerical procedures for calibration and path generation of the heavy-tailed stochastic processes, in particular, to model the dynamics of the equity markets. I developed an APT-style two-factor model to separate market-wide effects from the idiosyncratic shocks of the stocks under consideration. For the modelling of market-wide effects Vasicek and Heston models were used, while idiosyncratic returns were studied under the framework of the Merton Jump-diffusion process. All models were calibrated on historic data via Max-Likelihood estimation. I compared the accuracy of model calibration via Max-Likelihood and Method of Moments to the conclusion that the use of latter has led to a minor decrease in back-testing Log-Likelihood, while being highly computationally efficient.

Аннотация

В данной работе я предлагаю ряд численных методов для калибровки и генерации случайных процессов, основанных на тяжелохвостных распределениях, для анализа доходностей акций. В работе была использована двухфакторная модель, основанная на Теории арбитражного ценообразования, с помощью которой удалось отделить общерыночные эффекты от идиосинкратических шоков отдельных акций. Для моделирования риск-факторов были использованы модели Васичека и Хестона, а идиосинкратические шоки рассматривались в рамках прожково-диффузионной модели Мертона. Все модели калибровались с помощью Метода Максимального Правдоподобия на исторических данных. Сравнение качества оценок, полученных с помощью ММП и Метода Моментов, при помощи бэк-тестирования показало, что использование Метода Моментов незначительно снижает качество калибровки модели, однако приводит к кратному росту скорости самой калибровки.

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1 Introduction

Stochastic processes has been actively used for the modelling the returns on the financial markets for over 50 years, with the first attempts dating back to the beginning of the 20th century. However, most contemporary models used by the industry professionals and academics alike assume the Normality of the returns, which is contradicted a number of stylized facts regarding financial markets, most importantly, excess kurtosis and leftwards skewness of the returns distributions.

The aim for this work is to develop a framework with the help of the tools of stochastic calculus and numerical methods to account for these stylized facts within the context of equity markets. In particular, I am interested in the modelling of dividend gaps as the kind of rapid changes in stocks' prices (so-called jumps).

Two focal points of my work are model calibration and path generation. The first refers to the algorithm of selecting the parameters that most accurately reflects the dynamics of the process (in this case, stock returns). The latter is the procedure of turning continuous stochastic process into a discrete one in order for computers to be able to simulate it. The important consideration for me is the efficiency of these two procedures. I provide a number of tests and simulations that allow me to compare both accuracy and computational efficiency of different approaches considered.

Finally, I focus on writing a number of universal scripts that can be used for the analysis of the financial data, as well as calibration and simulation of random processes of different classes.

2 Literature Review

Before moving on to the practical part of my work, it is important to cover the theoretical foundations of stochastic processes, current best practices of their uses in the real-world setting, as well as key papers that try to take into account the empirical effects, not reflected in the currently used models. To do that, I will devote three separate subsections to these topics, covering most significant research done in the areas of stochastic calculus and financial mathematics, empirical findings regarding heavy-tailed distributions in finance, and the theory of jump-diffusion stochastic processes.

2.1 The history of stochastic calculus and financial mathematics

The questions regarding probabilities have been on great mathematicians' minds for centuries, dating back to the works of Pascal [8] and Bernoulli [30]. The first attempt at statistical inference was done by John Graunt [37], which was influenced by Christiaan Huygens [12], who developed the notion of expected value. Pierre Simon de Laplace [50] stated the addition and multiplication rules of probability, as well as applied them to the number of problems revolving around Bernoulli process, which was actively studied by the mathematicians in the beginning of the 19th century.

The first applications of Probability Theory were often related to physics problems, for example, modelling a closed system of gas filled container. Such problems together formed a discipline of statistical mechanics. The first work that extensively used the notion of randomness to model physical system was two papers by James Clerk Maxwell: [31] and [32] that laid the foundations of Kinetic theory of gases. Ludwig Boltzmann [41] extended the theory of Maxwell to solids and liquids, and Josiah Willard Gibbs [38] used the same principles in his work on physical optics. All these papers heavily influenced Albert Einstein, who performed the first statistical physics analysis of Brownian motion in his 1905 paper [3]. In 1911, Jean Baptiste Perrin [33] empirically proved the theoretical findings of Einstein, ending a century-long debate with the confirmation of the Atomic Theory.

However, five years before the groundbreaking work of Einstein, Louis Bachelier in his now-famous thesis "The Theory of Speculation" [6] used stochastic calculus in application to finance. He introduced the idea of the randomness of prices on the financial markets and used it to derive the fair

value of the European option. Bachelier's ideas were mostly ignored by the general public, however, his research later played a major role in the formation of modern-day financial markets.

At the same time, the work of defining the basic postulates of probability theory was coming to an end. The milestone work of A. N. Kolmogorov [1] formally drew a distinction between mathematics and probability, yet integrating the two into a coherent system.

The next breakthrough came in 1951, when Japanese mathematician Kiyoshi Ito published his most famous work [40], which for the first time presented a principle that would later become known as Ito's Lemma. Around the same time, Harry Markowitz published his most influential paper [28], which formed the foundation of Modern Portfolio Theory. This led to the development of modern-day CAPM by William Sharpe in his work [63] and its extensions - general Arbitrage Pricing Theory (APT), introduced in [56] by Stephen A. Ross, and its concretizations: 3- and 5-factor Fama-French models, introduced in [21] and [22].

However, the paper that truly formed the modern financial mathematics was "The Pricing of Options and Corporate Liabilities" [24] by Fischer Black and Myron Scholes, followed by "A Rational Theory of Option Pricing" [53] from Robert Merton, which independently derived the price of the European Option under the Geometric Brownian motion. A Feynman–Kac Theorem, which was proposed in the 1940s, linked the two distinct approaches when working with Options' fair value - calculating the expected discounted risk-neutral payoff, and solving the famous Black-Scholes PDE:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0 \tag{1}$$

After the 1970s, thanks to the exponential growth of the options and derivatives markets, the field of financial mathematics diverged into dozens of distinct sub-fields and approaches. Many researchers tried to relax the assumptions of the original Black-Scholes model by introducing stochastic interest rates ([47] by Oldrich Vasicek, [36] by John C. Cox et al. and [2] by Alan D. White and John Hull) and stochastic volatility ([57] by Steven L. Heston, [35] by John C. Cox and [49] by Patrick S Hagan et al). Other notable fields of study include CVA (Credit Valuation Adjustment), which emerged after the crisis of 2007-8, continuous research into pricing derivatives with PDEs and Monte-Carlo

methods, such as its variance reduction (which is especially important for pricing more complex and path-dependent financial products).

Finally, it has been empirically shown that asset returns do not exhibit a perfect Normal distribution, but have so-called "heavy tails" - extreme events tend to occur more frequently than predicted by the Gaussian Distribution, and their effect is more significant. The study of heavy-tailed distributions and the approach to account for them in the pricing of the financial securities is the main focus of this paper.

2.2 Heavy-tailed distributions in empirical finance

It has been long observed that price movements on financial markets do not follow a Normal distribution, instead having "heavy tails", which make extreme events more likely to occur. Even Louis Bachelier in his most famous work [6] discussed how not all price movements are born out of the Gaussian distribution. The first rigorous empirical study of the distribution of the financial markets was done by Benoit Mandelbrot in [7]. He successfully showed that price returns do not follow a Normal distribution, instead having long stretched-out tails and high density around the mean:

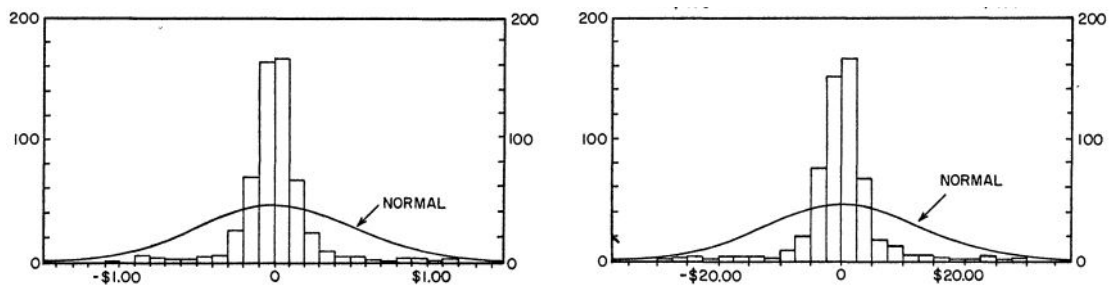


Figure 1: 5- and 10-month returns against Normal PDF. Source: B. Mandelbrot's work

On a more technical note, the techniques that can be used for finding the existence of heavy tails are numerous, with the most basic being demonstrated above - plotting a histogram of empirical returns against the theoretial pdf of the Normal distribution. If two graphs diverge significantly, the distribution of returns cannot be considered Normal. Another approach is so-called Q-Q plot: the idea is to graph a scatter plot with 100 points, where each point is characterised by the quantiles of the empirical and Normal distribution. If the straigt line is formed, empirical distribution is believed

to be Normal. Below is the illustration of the Normal distribution tested with Q-Q plot:

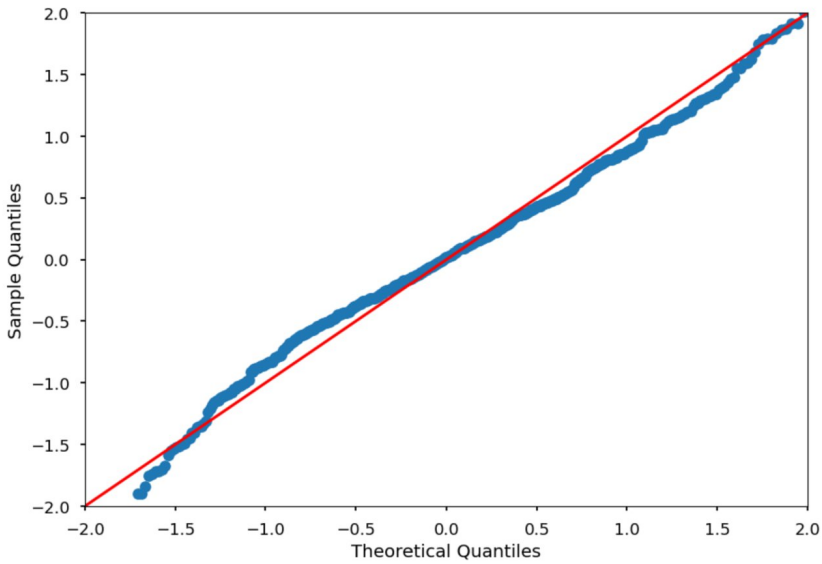


Figure 2: Q-Q plot of the Normal distribution generated by the author

However, these approaches are not quantitative and rely on human perception. In order to get rid of this drawback, two metric were proposed: it is possible to calculate the kurtosis of a sample of the empirical returns with the formula

$$\kappa = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^4}{\left(\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2\right)^2} \tag{2}$$

The obtained number is then tested for Normality with the method proposed in [23] by F. J. Anscombe and William J. Glynn. Then the usual conditions for the rejections of Normality, based on confidence level, apply.

Finally, it is possible to use Kolmogorov–Smirnov test, which allows to check, whether a sample is taken from a particular distribution, which is the most formal approach to the distribution testing.

Similar research regarding the presence of heavy tails was done by a vast array of authors throughout the last 50 years, with findings indicating that Foreign Exchange ([48] by P. Hartmann et al., [44] by Michel M. Dacorogna et al. and [43] by Marat Ibragimov et al.), equities ([64] by Zi-Yi Guo, [4] by

Almira Biglova et al. and [45] by Namwon Hyung and Casper G. de Vries) and commodities ([39] by Jullavut Kittiakarasakun and [46] by Nicolas Merener) all exhibit heavy-tailed distributions of returns.

The new wave of interest in the distributions that account for extreme events on the financial markets emerged after the "Black Monday" of 1987, Dot-Com bubble and 2007-8 Subprime mortgage crisis. As a result, new approach to heavy-tailed modelling emerged - it is possible to reject the use of Gaussian distribution altogether and replace it with some other distribution, which has higher kurtosis. Notable works in this field include [59] by Tim Bollerslev, [42] by M. Beine et al. and [60] by Timotheos Angelidis et al., which focus on applying Student's t-distribution when modelling returns on specific assets; [20] by Ender Su and Thomas W. Knowles, which makes use of the Generalized Error Distribution (GED); and [9] by Boris Podobnik et al., centered around the Power law distribution.

However, the approach that I want to focus on is jump-diffusion processes. The basic idea is simple: have a Geometric Brownian motion be the main process, with infrequent "jumps" - large instantaneous changes in processes value (in other words, abnormally high or low return for a given period).

The first and arguably most famous jump-diffusion stochastic process was developed by Robert C. Merton in his work [52], published in 1976. As will be discussed below, it assumes the Normality of the Jump amplitude. A group of models called Basic Affine Jump Diffusion (basic AJD) became popular in the tasks of corporate and retail credit, and short-rate modelling (notable works which use these models include [14] by Darrell Duffie and Nicolae Gârleanu and [18] by David S. Bates). Currently, there is active research focusing on correlated diffusion and jumps (for reference, see [65] by Yihren Wu and Majnu John).

2.3 Jump-diffusion stochastic processes

In order to model jump-diffusion stochastic processes on the real-world data, I had to study the current approaches used in this field. The typical approach is the one by Merton, which is described by the Stochastic Differential Equation:

$$dS(t) = S_t\mu_d dt + S_t\sigma_d dW_t + S_t J(Q)dP_t \tag{3}$$

Where dP_t has a distribution of

$$dP_t \sim p_k(\lambda dt) = P(dP(t) = k) = e^{-\lambda dt}\frac{(\lambda dt)^k}{k!} \tag{4}$$

with $k \in [0, \infty]$. In other words, dP_t is the distribution of the number of jumps that occurred during an infinitesimally small time period dt. Note, that the Probability Density Function of that distribution is almost identical to the traditional PDF of Poisson distribution (which is given by $e^{-\lambda}(\lambda)^k/k!$), with the only difference of it being "squished" by some constant dt, reflecting the shortened time interval.

The distribution above can be transformed to a log-return form with an Ito's Lemma:

$$d(\ln(S_t)) = (\mu_d - \sigma_d^2/2)dt + \sigma_d dW_t + \ln(1 + J(Q))dP_t \tag{5}$$

With $\ln(1 + J(Q)) = Q \sim \mathcal{N}(\mu_j, \sigma_j^2)$ and average jump $J(\bar{Q}) = e^{\mu_j + \sigma_j^2} - 1$. Note, that all three sources of randomness in the model $(dW_t, J(Q), dP_t)$ are independent. However, in [55] S.G. Kou additionally points out that $J(Q)$ is conditional on dP_t being > 0.

The solution to this SDE is:

$$S_t = S_0 e^{(r-\lambda\bar{Q}-\sigma_d^2)t+\sigma W_t}\prod_{i=1}^{P_t}(1 + J(Q_i)) = S_0 e^{(r-\lambda\bar{Q}-\sigma_d^2)t+\sigma W_t+\sum_{i=1}^{P_t} Q_i} \tag{6}$$

To move forward I apply the technique from [25] and [26] by Floyd B. Hanson and J. J. Westman to write a PDF of $d\ln(S_t)$, where $\mu_{ld} = \mu_d - \sigma_d^2/2$. Importantly, in these papers, there is a significant alteration made to the original Merton model: $J(Q)$ is log-uniform, not log-normal, with Q being uniformly distributed on the interval $[Q_a, Q_b]$, with the density $\phi_Q(q) = U(q, Q_a, Q_b) = 1/(Q_b - Q_a)$. Obviously, its moments are $\mu_j = (Q_a + Q_b)/2$ and $\sigma_j^2 = (Q_b - Q_a)^2/12$.

Then the Probability Density Function of the returns of a process is given by:

$$\phi_{d\ln(S_t)} = p_0(\lambda dt)\phi^{(n)}(x, \mu_{ld}dt, \sigma_d^2 dt) + \sum_{k=1}^{\infty} p_k(\lambda dt)\frac{\Phi^{(n)}(x - kQ_b, x - kQ_a, \mu_{ld}dt, \sigma_d^2 dt)}{k(Q_b - Q_a)} \tag{7}$$

With $p_k(\lambda dt)$ being the probability of k jumps occurring during dt and $\Phi^{(n)}(x, y, \mu_{ld}dt, \sigma_d^2 dt)$ being the Normal distribution on $[x, y]$, defined as:

$$\Phi^{(n)}(x, y, \mu_{ld}dt, \sigma_d^2 dt) = \int_x^y \phi^{(n)}(z, \mu_{ld}dt, \sigma_d^2 dt)dz = \int_x^y \frac{exp(-(z - \mu_{ld}dt)^2/(2\sigma_d^2 dt))}{\sqrt{2\pi\sigma_d^2 dt}} \tag{8}$$

Therefore, the expression for PDF of $d(\ln(S_t))$ has quite a intuitive meaning: the first term represents the case if there is no jump occuring during dt. Therefore, the change in the stochastic process value is the Normal variable with mean $\mu_{ld}dt = (\mu_d - \sigma_d^2/2)dt$ and variance $\sigma_d^2 dt$. The second term represents a sum of all jumps that occur during dt (with total being equal to k). It has the probability of k'th jump multiplied by the jump size, defined as the integral of Normal distribution that falls inside the Uniform distribution interval, normalized through the division by the non-zero interval of the uniform distribution Q.

As in reality in financial markets dt is not observed (as it is infinitesimal of time), but Δt can be measured (for example, using one-day time intervals), ignoring $O((\Delta t)^2)$ we can transform our PDF to be asymptotically approximated as:

$$\phi^{(jd)}(x) \approx (1 - \lambda\Delta t)\phi^{(n)}(x, \mu_{ld}\Delta t, \sigma_d^2\Delta t) + \lambda\Delta t\frac{\Phi^{(n)}(x - Q_b, x - Q_a, \mu_{ld}\Delta t, \sigma_d^2\Delta t)}{Q_b - Q_a} \tag{9}$$

Finally, if we assume that time intervals Δt are small enough, we note that realistically, only 0 or 1 jumps can occur during it (usually 0, with rare exceptions of a single jump occurring). Given that all coefficients are constant, we can transform the expression above to an even simpler form:

$$\Delta ln(S_t) = \int_t^{t+\Delta\tau} dln(S_\tau) \sim \mu_{ld}dt + \sigma_d\Delta W_t + Q\Delta P_t \tag{10}$$

Which, when we consider risk-neutral setting, results in:

$$\Delta ln(S_t) \sim (\mu_d - \lambda\mu_j - \sigma_d^2/2)dt + \sigma_d \Delta W_t + Q\Delta P_t \tag{11}$$

This construction works even when other distributions (for example, Normal, like in Merton model) are considered. Authors of the papers, mentioned above, argue that using Uniform distribution is actually desirable to the use of Normal one. If Normal distribution is used, two sources of randomness (dW_t and Q) will have the same distribution. Then the situation may occur, when the calibrated parameters for the mean and variance will turn out to be close between these two distributions, which would nullify the effects of the jump term, eliminating the leptokurtosis of the model price returns. The original Merton paper avoids this problem by selecting the parameters of Q not by some optimization algorithm, but by hand-picking "jump points" from the price data. Meanwhile, Uniform distribution "forces" heavy tails ontop of Normal distribution. However, as I demonstrate in this work, this suspicion is unfounded, as it is possible to calibrate Merton process into two Normal distributions with distinctly different parameters.

Please note, that I alter the original notations from the papers to ensure consistency between different sources used. The model above has 5 parameters: deterministic trend born out of GBM μ_d, GBM diffusion standard deviation σ_d, jump frequency (rate) λ, jump amplitude mean μ_j and jump amplitude standard deviation σ_j, which will be the calibration parameters in case of Merton model. However, if the log-uniform model is considered, μ_j and σ_j can be expressed through Q_a and Q_b.

Such model architecture allows researchers to account not only for sudden surges and crashes, but also the skewness of returns (large negative outliers are larger than positive ones) and leptokurtosis (positive excess kurtosis, "fat tails"), which, as discussed above, are all features of returns present on the real financial markets.

3 Proposed methodology

As my paper focuses on dividend gaps, it is crucial to model idiosyncratic risks, associated with specific stocks. For that, it is required to separate idiosyncratic (stock-specific) risk from the systemic one. I propose to do that through Arbitrage Pricing Theory (APT), introduced in 1976 by Stephen A. Ross [56]. In it, observable returns are a sum of two values: systemic and idiosyncratic components. Therefore, I argue, it is possible to get pure idiosyncratic returns by subtracting market-wide effects from the observable returns:

$$R_{observable} = R_{market} + R_{idio} \Rightarrow R_{idio} = R_{observable} - R_{market} \tag{12}$$

Observable returns are easily available. However, in order to model R_{market} it necessary to select market-wide risk factors, associated with all traded stocks.

3.1 Choosing risk factors for APT

In order to apply APT to get residual (idiosyncratic) returns, we must run a regression of R_{it} - the observable returns of a stock i at time t, on various risk factors:

$$R_{it} = \beta_0 + \sum_{j=1}^{N} \beta_i f_{jt} + \epsilon_{it} = \beta_0 + \sum_{j=1}^{N} \beta_i f_{jt} + R_{idio} \tag{13}$$

Where f_{jt} is the value of a specific risk factor j at the time t, β_i is the sensitivity to a risk factor i, β_0 is the constant term of the model (which I expect to be equal to risk-free rate r_f) and ϵ_{it}, which is the time series of the residual returns of the stock i.

Now, I have to find the optimal list of risk factors that I will regress, according to the model described above. We can start by examining the approach chosen by the original paper [56] that introduced Arbitrage Pricing Theory. Stephen A. Ross proposed the use of market excess return $R_{Mt} - r_{ft}$ as a risk factor. This choice makes sense for this model, as the time series of S&P500 closing values does not exhibit stationarity, but its first difference (returns) does:

Time series	Test statistic of ADF test	P-value	Result
S&P500 closing values	-2.16	0.22	Not stationary
S&P500 returns	-27.32	0.00	Stationary

Therefore, using returns of S&P500 as a market proxy is preferable.

However, another important factor necessary to consider is the volatility of the overall market. It can be estimated using VIX index, publicly available on information platforms such as Yahoo finance. This index represents the expected volatility of S&P500 index in the next 30 days from implied volatility of the Options on the SPX and SPXW indices (which means it is forward-looking and can be modelled with traditional volatility models). For it, differencing is not necessary, as the original time series is already stationary:

Time series	Test statistic of ADF test	P-value	Result
VIX closing values	-3.70	0.00	Stationary
VIX returns	-16.10	0.00	Stationary

Finally, there is another risk factor that I implicitly reference in this setup, and it is the risk-free rate r_{ft}. It will have its own dynamic, based on stochastic process, and it was already accounted for in my formulation of asset's excess return for both individual asset and broad market.

Therefore, the APT-based model takes form:

$$R_{it} - r_{ft} = \beta_0 + \beta_1(R_{Mt} - r_{ft}) + \beta_2 VIX_t + \epsilon_{it} \tag{14}$$

Given that we fix sensitivities to risk factors (betas) in time and we can observe all variables in the regression besides residual term ϵ_{it}, which we can easily express it as:

$$\epsilon_{it} = R_{it} - r_{ft} - \beta_0 - \beta_1(R_{Mt} - r_{ft}) - \beta_2 VIX_t = R_{it} - \beta_0 + (\beta_1 - 1)r_{ft} - \beta_1 R_{Mt} - \beta_2 VIX_t \quad (15)$$

Using that model, we calculate the residual time series of the number of stocks.

3.2 Statistical properties of the residual time series of returns

For the purposes of this analysis, I have chosen a range of stocks, which time series of returns I have analysed using a methodology proposed above. The stocks include AAPL, MSFT, BA, LLY, GME, NVDA, NFLX, GOOG and CRM, and time period chosen for the analysis is between January 1st, 2021 and January 1st, 2024. I begin the analysis of these idiosyncratic returns by looking at their plots and contrasting them with the observed returns, that also contain systemic risk. Below are the graphs of such residual returns for three stocks from my sample:

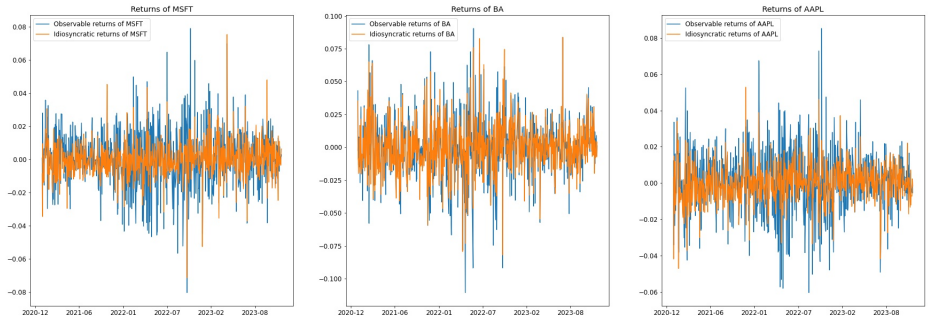


Figure 3: Observable and residual (idiosyncratic) returns of MSFT, BA and AAPL

The visual analysis of the new time series suggests, that idiosyncratic returns exhibit a significantly lower volatility, compared to its observed version. However, large changes in stocks' prices are usually at least partially explained by the idiosyncratic factors (in simple terms, when blue plot swings heavily, it usually coincides with the large changes in orange plot).

Furthermore, we can study the statistical properties of these new time series: to begin with, I start with examining the first four moments of the obtained samples:

Residual returns	Mean	StDev	Skewness	Kurtosis	Kurtosis of observable returns
AAPL	0.0000	0.0105	-0.0562	2.9401	1.5997
MSFT	0.0000	0.0110	0.2156	6.8075	1.6293
BA	0.0000	0.0193	0.3146	2.5506	2.0360

Importantly, idiosyncratic returns do not exhibit a statistically significant mean (market drift), and have higher excess kurtosis compared to observable returns. This indicates that residual returns have heavier tails compared to observable ones, in other words, extreme events have larger magnitude. Therefore, in order to explain such phenomenon and construct a more accurate model, it is necessary to step away from Geometric Brownian Motion, which assumes the Normality of returns. One possible substitute are Jump-diffusion processes, discussed above, and which I am going to use moving forward.

3.3 Construction of Jump-diffusion process

I am going to base my work on the Merton Jump-diffusion model, described above. Before, I propose the method of its construction, it is important to write down a simple proof used for the simplification of calculations when working with the model:

As described in the literature review section of this paper, the risk neutral return of the process is described as:

$$\Delta ln(S_t) \sim (r - \lambda\mu_j - \sigma_d^2/2)dt + \sigma_d \Delta W_t + Q\Delta P_t \tag{16}$$

Where ΔP_t is an indicator, whether the jump has occurred during Δt. The probability of jump occurring can be found through Poisson distribution. For example, assume that the analysis on the daily data is performed, with jumps on average taking place every 100 trading days. This implies the probability of a jump:

$$P(\Delta P_t = 1) = 1 - p_0(\lambda \Delta t) = 1 - p_0(1/100) \approx 0.01 \tag{17}$$

Furthermore, given small time intervals Δt and low jump frequency, it is reasonable to approximate the probability of a jump as:

$$P(\Delta P_t = 1) \approx \lambda \tag{18}$$

Which we can prove by plugging the expression for the pdf of Poisson distribution:

$$P(\Delta P_t = 1) = 1 - \frac{\lambda^k e^{-\lambda}}{k!} = 1 - \frac{(1/100)^0 e^{-1/100}}{1} = 1 - e^{-0.01} \tag{19}$$

And then by Taylor series:

$$e^{-0.01} \approx e^0 + e^0(0 - 0.01) + O(x^2) = 1 - 0.01 + O(x^2) \tag{20}$$

Then I plug it back into the probability of a jump occurring:

$$P(\Delta P_t = 1) = 1 - e^{-0.01} \approx 1 - 1 + 0.01 = 1 - 1 + \lambda = \lambda \tag{21}$$

In means, that we can approximate the probability of the jump occurring during a small time period Δt as λ, which I am going to use moving forward. Now, I propose the following methodology for the generation of the Merton Jump-diffusion process:

1. Generate an array with dimensions (n_{steps}, n_{paths}) (number of steps in the simulation and number of independent paths within each simulation) and fill it with Bernoulli random variables, which have a parameter λ. Assign it to a variable B.

2. Generate Normal random variables with parameters (μ_d, σ_d) in the array, which also has a size of (n_{steps}, n_{paths}). This will be a non-jump part of the stochastic process. Assign it to a

variable D.

3. Generate one more array of size (n_{steps}, n_{paths}) with Normal random variables, but this time with parameters (μ_j, σ_j). This is the jump term of the process. Assign it to a variable J.

4. Create a new variables, which will be an array with dimensions (n_{steps}, n_{paths}), equal to $T = D + J \times B$. In other words, in case a jump occurs (which has a probability of happening $= \lambda$) add it to the regular "non-jump" process.

5. Perform cumulative summation of the array T across individual trajectories. Shift the entire array by some constant, that reflects the starting values of the process (usually, it is the last price from the historic time series). These are the paths of the simulation. Now, they can be saved or visualized using a graph

Furthermore, it can easily be noted, that the PDF of this process equals to:

$$f^T(\lambda, \mu_d, \sigma_d, \mu_j, \sigma_j) = \frac{1}{1+\lambda} f^J(\mu_j, \sigma_j) + \frac{\lambda}{1+\lambda} f^D(\mu_d, \sigma_d) \tag{22}$$

Where $f^T(\lambda, \mu_d, \sigma_d, \mu_j, \sigma_j)$ is the PDF of the Jump-diffusion process, $f^J(\mu_j, \sigma_j)$ is the PDF of the Jump process, and $f^D(\mu_d, \sigma_d)$ is the PDF of the regular diffusion process.

Therefore, the likelihood for Jump-diffusion process can be defined, given an array of observed values $x = [x_1, x_2, \cdots x_n]$:

$$Likelihood = \prod_{i=1}^{n} f^T(x_i, \lambda, \mu_d, \sigma_d, \mu_j, \sigma_j) \tag{23}$$

See the section below for the approach I used for determining the parameters of $f^T(\lambda, \mu_d, \sigma_d, \mu_j, \sigma_j)$. On the final note in this section, the same approach applies to a Multivariate distribution, with μ_d and μ_j becoming vectors of length equal to the number of dimensions in the distribution, and σ_d with σ_j being a covariance matrices of the stocks' returns.

4 Paths simulation and model calibration

In this section I am going to cover the methodology used for the simulation of paths and model calibration. These problems are a part of a branch of mathematics called numerical methods, which tries to find accurate, yet efficient algorithms to solve mathematical problems. In the context of my work, these two problems are crucial to solve in order to achieve any meaningful results.

4.1 Discretization and path generation

The first problem revolves around finding the algorithm to transition from continuous differential equations to the simulation of stochastic processes on the discrete grid. The reason for it is trivial: computers cannot work with truly continuous variables, therefore, they are unable to simulate the true differential equations (both regular and stochastic). However, if the differential equations can be approximated on the discrete grid, it is possible to analyse them and make predictions about their future trajectories (this aspect is particularly important to me, as I need to be able to simulate stochastic processes to the future).

The most well known algorithm for the approximation of differential equations is Euler method. It works in the following way:

1. Consider the case when we have a differential equation of the form: $y'(t) = f(t, y(t))$.
2. Let h be some real number, that will represent a time shift between previous and next point of the approximation (it can be considered the degree of the finiteness of the discrete grid)
3. Given that we can approximate $y(t+h)$ with a first-degree Taylor series, $y(t+h) \approx y(t)+y'(t)h$
4. Using new time $t+h$ and new approximated value of a function $y(t+h)$, it is possible to find new approximate derivative $y'(t+h) = f((t+h), y(t+h))$.
5. Now, the procedure can be repeated, with new values of time, the function and its derivative.

Furthermore, this method generalizes to ODEs of an order > 1. The procedure can be written down in similar terms: given explicit differential equation $y^{(N+1)}(t) = f\left(t, y(t), y'(t), \cdots, y^{(N)}(t)\right)$:

$$\begin{pmatrix} y(t+h) \\ y'(t+h) \\ \vdots \\ y^{(N)}(t+h) \end{pmatrix} = \begin{pmatrix} y(t) + y'(t)h \\ y'(t) + y''(t)h \\ \vdots \\ y^{(N)}(t) + y^{(N+1)}(t)h \end{pmatrix} \tag{24}$$

Analogously to the ODE with order = 1, the procedure is then repeated with new values of time, the function and its derivatives.

Generally, Euler method is both inaccurate and inefficient, however, due to its simplicity, it is still actively used in many applications. There also exists an extension of this method to the Stochastic Differential Equations: it is called Euler–Maruyama method. The method is similar to the approach discussed above: given a stochastic differential equation $dX_t = a(t, X_t)dt + b(t, X_t)dW_t$ and time increment h, it is possible to estimate $X_{t+h} = X_t + a(t, X_t)h + b(t, X_t)W_h$, where $W_h \sim N(0, h)$.

On a computational side, if the drift term $a(t, X_t)dt$ and diffusion term $b(t, X_t)dW_t$ do not depend X_t, it is possible to simulate all increments at once, and then sum them up to obtain random paths for the process.

For example, considering Algebraic Brownian motion, described by the SDE:

$$dX_t = \mu dt + \sigma dW_t \tag{25}$$

This process can be simulated by generating an array of random variables with length of total number of steps in the simulation (I am going to use this word to describe the process, analogous to the approximation of ODEs) and height equal to the number of paths to be generated, each following $N(\mu dt, \sigma^2 dt)$, and then performing a cumulative sum of them across each path. For the example of the script that performs such generation, I encourage you to see the attached archive with .py files.

Otherwise, the simulation has to be done sequentially, through the looped script. For example, the Ornstein–Uhlenbeck process is described by the SDE:

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t \tag{26}$$

Its forms are used to describe volatility in Heston model short rates in Vasicek model (they will be examined in details below). As the next value of the process depends on the previous one, it is impossible to simulate all the increments at once - it has to be done sequentially. The algorithm of simulating such process is to loop through each point in time on a discrete grid using the previous value of a process in every path as input for the drift term. The resulting process has a property of mean reversion - a desired trait for both volatility and interest rate simulation.

4.2 Model calibration

With the discretization, it becomes possible to simulate the paths of stochastic processes. However, we should also determine such array of parameters, passed into the model, that make the best prediction of the possible future paths of a process. Most researchers agree that the past data should be used to determine these parameters, which are then passed for simulation of future paths.

The first challenge in this process is finding a way of measuring model's "goodness". The approach I chose for this paper is Max-Likelihood Estimation. It determines a set of parameters, for which the likelihood (or more specifically, log-likelihood) is maximized. This approach works in the following way:

1. A time series of historic data (more generally, any sample) is taken as an input. It is stored as a vector $x = [x_1, x_2, \cdots, x_n]$
2. The assumption has to be made regarding the true distribution of the sample (for example, it can be Normal, Exponential, Geometric - the choice depends on the nature of the problem). The chosen distribution is required to have explicit and known Probability Density Function (PDF, f), which takes a known set of parameters θ.
3. Plug the empirical sample, as well as parameters vector, into the PDF to get $f_i = f(x_i, \theta)$
4. Evaluate log-likelihood, defined as $L = \sum_{i=1}^{n} \log(f_i)$

5. Use an analytical or numerical algorithm to maximize L with respect to a parameter vector θ. The resulting optimal parameters θ^* are used for paths simulation.

This approach is widely known, and has proven its effectiveness in many applications, such as Machine Learning. However, it is a well-known fact that MLE is biased. For example, it tends to underestimate the variance of a distribution. Furthermore, when working with time series, it may be desirable to give recent observations more weight in parameter calibration. The approach known as Weighted Likelihood estimation (WLE), first proposed by Thomas A. Warm in [58], suggests that it is better to exponentially weight the likelihoods:

$$L = \frac{1-\eta}{1-\eta^n} \sum_{i=1}^{n} \eta^{i-1} \log(f_{n+1-i}) \tag{27}$$

Where η is the sensitivity parameter, which is usually set to 0.95 for daily data (in most sources the sensitivity is notated as λ, but in order to avoid confusion with the probability of a jump, I chose a different notation)

The similar idea was later proposed in JP Morgan's RiskMetrics™ in the technical document [29]. It was confirmed in [51] by Pilar Abad and Sonia Benito that the approach of weighted historical simulation for modelling Value-at-Risk performs comparatively better against the baseline of regular historical simulation and parametric models.

Therefore, WLE has lower bias in combination with solid economic and econometric interpretation, however, I decided not to use it in my optimization procedures, as it will make the output impossible to compare with estimates, obtained via other methods.

Now, the question arises, how to optimize this likelihood function (find the best set of parameters)? There is a number of methods to solve this problem, which are discussed in details in the following pages.

4.2.1 Analytical solution

Then there are two broad ways one can approach such problem: the first one is to find an analytical solution, which can be done by finding gradient of the partial derivatives of the Log-likelihood

function with respect to the every parameter in θ and selecting set of points which make it equal to zero.

$$\nabla f(\theta) = \begin{bmatrix} \frac{\partial f}{\partial \theta[1]}(\theta) \\ \frac{\partial f}{\partial \theta[2]}(\theta) \\ ... \\ \frac{\partial f}{\partial \theta[n]}(\theta) \end{bmatrix} = 0 \tag{28}$$

After that, it is necessary to check if the Hessian (a matrix of the second partial derivatives) is negative definite for every point in this set.

$$x^T \mathcal{H} x = x^T \begin{bmatrix} \frac{\partial^2 f}{\partial \theta[1]^2}(\theta) & \frac{\partial^2 f}{\partial \theta[1]\partial \theta[2]}(\theta) & ... & \frac{\partial^2 f}{\partial \theta[1]\partial \theta[n]}(\theta) \\ \frac{\partial^2 f}{\partial \theta[2]\partial \theta[1]}(\theta) & \frac{\partial^2 f}{\partial \theta[2]^2}(\theta) & ... & \frac{\partial^2 f}{\partial \theta[2]\partial \theta[n]}(\theta) \\ ... & ... & ... & ... \\ \frac{\partial^2 f}{\partial \theta[n]\partial \theta[1]}(\theta) & \frac{\partial^2 f}{\partial \theta[n]\partial \theta[2]}(\theta) & ... & \frac{\partial^2 f}{\partial \theta[n]^2}(\theta) \end{bmatrix} x < 0 \tag{29}$$

For all non-zero x in $\mathbb{R}^n$.

However, given the amount of data and complex nature of both Log-likelihood function and model returns, it is more reasonable to use numerical methods instead. There are many numerical minimization techniques, most of them are available within optimization libraries in a number of programming languages (in my work I mostly used Python's scipy.optimize library).

4.2.2 Newton-CG method

The first and arguably most well known algorithm is Newton method with conjugate gradient. The most basic form of Newton method notes, that the second-order Taylor expansion of a single-variable function f around x_k is:

$$f(\theta_k + \Delta\theta) \approx f(\theta_k k) + f'(\theta_k)t + \frac{1}{2} f''(\theta_k)t^2 \tag{30}$$

Then new x_{k+1} is chosen to minimize that Taylor approximation. As approximation is parabola, its minimum is $t^* = -f'(x_k)/f''(x_k)$. Then new point of $x_{k+1} = x_k - f'(x_k)/f''(x_k)$ is chosen, and procedure is repeated. Obviously, the algorithm requires some initial "guess" x_0 to work.

To generalize Newton method to the higher dimensions, it is necessary to replace $f'(x_k)$ with $\nabla f(\theta)$ and $f''(x_k)$ with the inverse of the Hessian matrix $[\mathcal{H}_f(\theta)]^{-1}$ to get the iterative next point:

$$\theta_{k+1} = \theta_k - [\mathcal{H}_f(\theta_k)]^{-1} \nabla f(\theta_k) \tag{31}$$

As one can easily notice, this method has three major problems:

1. It requires an initial point set by the user
2. It works well only with convex functions
3. Most importantly, it requires a calculation of the inverse of a matrix, which is, even with the best algorithm, $O(n^3)$, which is considered incredibly slow and inefficient (however, if n - the number of parameters to optimize, is relatively small, the algorithm still can be fast)

However, there is a way to use this algorithm without the third limiting factor. It is possible to approximate a vector h, that is a solution to

$$[\mathcal{H}_f(\theta)]h = -\nabla f(\theta) \tag{32}$$

One of the iterative methods of finding that vector is conjugate gradient (which is indicated by CG in the Newton-CG method) that makes possible to find solutions to the systems of linear equations. However, one important condition on which it is based is that original Hessian matrix must be Positive definite (that is why the maximization problem is turned into a minimization one by the multiplication of the target function by (-1) - to ensure the positive definiteness of the Hessian).

Finally, a number of other approaches to making this algorithm faster and more efficient were developed, most notably, [61] by Trevor J. McDougall and Simon J. Wotherspoon, [62] by Tuyen Trung Truong at el. and [13] by D. C. Sorensen.

4.2.3 Broyden–Fletcher–Goldfarb–Shanno method

Another method could also be used: its name is Broyden–Fletcher–Goldfarb–Shanno algorithm (BFGS for short). It is also an iterative algorithm, actually, it is classified as Quasi-Newton, because by construction it is quite similar to the method discussed above.

Consider an approximation B_k of the inverse Hessian matrix $\mathcal{H}_f(\theta)$. Then we need to obtain a direction vector p_k by solving

$$B_k p_k = -\nabla f(\theta) \tag{33}$$

Then line search is performed to find an optimal step size α_k taken in the direction p_k. In its exact form line search is a problem of:

$$\alpha_k = \operatorname{argmin} f(\theta_k + \alpha p_k) \tag{34}$$

Find new point:

$$\theta_{k+1} = \theta_k + \alpha_k p_k \tag{35}$$

Then we set an intermediary measure of curvature:

$$y_k = \nabla f(\theta_{k+1}) - \nabla f(\theta_k) \tag{36}$$

And calculate the new approximate of inverse Hessian matrix:

$$B_{k+1} = B_k + \frac{y_k y_k^T}{y_k^T(\alpha_k p_k)} - \frac{B_k \alpha_k p_k (\alpha_k p_k)^T B_k^T}{(\alpha_k p_k)^T B_k (\alpha_k p_k)} \tag{37}$$

And then repeat the procedure until the convergence (which can be approximately checked by the condition $||\nabla f(\theta)|| \leq \epsilon$, in other words, gradient becomes approximately "flat").

The biggest downside of this method is that it requires curvature measure $\alpha_k p_k$ to be strictly positive and bounded away from zero. Additionally, this method is heavily memory-intensive. This downside is solved by the most popular modification of BFGS, called Limited-memory BFGS (or L-BFGS), proposed in [19] by Dong C. Liu and Jorge Nocedal. This version is most popular within most optimization libraries and software.

4.2.4 Gradient descent

Gradient descent is rightfully considered the most popular unconstrained optimization method in modern Machine Learning. As can be seen below, it is really similar by the procedure to the other methods discussed in this paper.

Lets start by selecting a starting point θ_0. Then iteratively update the "guess" of θ by:

$$\theta_{k+1} = \theta_k - \gamma \nabla f(\theta_k) \tag{38}$$

Where γ is a hyper-parameter, which represents learning rate, or the sensitivity of the updates of θ in response to the gradient. Note, that no condition on optimality of the exit has to be checked: as around local minimum the gradient converges to zero, changes in θ decrease in size, therefore, stability of an output is achieved. The user simply lets the algorithm run for some fixed number of iterations.

Note, however, that this method is incredibly prone to landing in saddle or "fake minimum" points, as it can arrive at such points and demonstrate convergence. To combat this downside, Stochastic Gradient Descent (SGD) was introduced. It partitions training set into multiple subsamples, that are independently used to train the model, therefore, resulting in a number of different solutions (some landing in "fake minimums", some - in real ones). Additional benefit of SGD is that as it can use only a subsample of the entire training set, it is computationally cheaper, which is compensated by the lower convergence rate.

4.2.5 Nelder–Mead method

Finally, it is important to cover Nelder-Mead method (also sometimes called downhill simplex method), introduced in [34] by John A. Nelder and R. Mead.

The algorithm is the following. Start by generating $n+1$ random points of θ and calculate values of the objective function in them: $f(\theta)$. Order function values:

$$f(\theta_1) \leq f(\theta_2) \leq ... \leq f(\theta_{n+1}) \tag{39}$$

Get rid of the largest value $f(\theta_{n+1})$ and calculate the θ_0 - the centroid of the remaining n points θ. Then compute reflected point:

$$\theta_r = \theta_0 + \alpha(\theta_0 - \theta_{n+1}), \alpha > 0 \tag{40}$$

If θ_r is not the best point among as set of θ (it does not have the lowest value), replace θ_{n+1} with it, and repeat the procedure until θ_r is the best. After that, compute an extended point:

$$\theta_e = \theta_0 + \gamma(\theta_r - \theta_0), \gamma > 1 \tag{41}$$

If $f(\theta_e) < f(\theta_r)$, replace θ_{n+1} with θ_e and repeat the entire process. Otherwise, replace θ_{n+1} with θ_r and repeat the entire process.

Then, if $f(\theta_r) < f(\theta_{n+1})$, compute the contracted point:

$$\theta_c = \theta_0 + \rho(\theta_r - \theta_0), 0 < \rho \leq 0.5 \tag{42}$$

If $f(\theta_c) < f(\theta_r)$, replace θ_{n+1} with $f(\theta_c)$ and repeat the entire process.

If, on the other hand, $f(\theta_r) \geq f(\theta_{n+1})$, compute the contracted point:

$$\theta_c = \theta_0 + \rho(\theta_{n+1} - \theta_0), 0 < \rho \leq 0.5 \tag{43}$$

If $f(\theta_c) < f(\theta_{n+1})$, replace θ_{n+1} with $f(\theta_c)$ and repeat the entire process.

Finally, replace all points, except "the best" θ_1 with

$$\theta_i = \theta_1 + \sigma(\theta_i - \theta_1) \tag{44}$$

Repeat this procedure until convergence is achieved. Cycle is stopped when the standard deviation of the points' coordinates in the sample falls below certain threshold.

A number of modification of the original procedure discussed above were developed over the years, for example, [54] by Russell R. Barton and John S. Ivey, Jr. focuses on reducing risks of inappropriate termination when simulating stochastic systems and [10] by C. J. Price which developed an approach to solve constrained optimization problems with Nelder-Mead method.

This concludes the section on the most popular numeric minimization procedures currently used in machine learning and optimization theory. I am going to focus on them when developing optimization algorithm for my stochastic models, while experimenting with their modifications. Generally, when testing various algorithms, I found out that Nelder-Mead yields the most stable results and is relatively high-performance even when used in high-dimensional optimization.

4.3 Computational infrastructure

The entire process of model calibration and path generation is performed using Python. I have written two scripts perform all the necessary calculations.

The first one is called data_analysis.py, and it contains a number of functions for the data parsing and engineering before model calibration can begin.

- **reader** function downloads the data from Yahoo Finance API. It accepts three parameters: list of stocks tickers for analysis, start and end dates for the historic window. The function

returns a list of stocks downloaded, a time series of these stocks, their returns (calculated as $\log(P_t/P_{t-1})$), values of S&P500 index and its returns, short term rates and volatility index VIX.

- **apt_betas** function runs a regression that estimates sensitivities of returns of each individual stock in response to changes in excess market returns and in VIX index. A dataframe with intercepts and betas of all stocks is returned.
- **correlation_matrix** builds a correlation matrix between the returns of the stocks chosen. It accepts a list of stocks, time series of stocks' returns and a boolean parameter, whether the correlation matrix should be visualized. The output is the (n, n) array, where n equals to the number of stocks passed, and, if chosen, a heatmap of this array.
- **covariance_matrix** is the same as previous function, but it returns a covariance matrix
- **idio_stock** takes the time series of stock returns, S&P500 returns and VIX, alongside their individual sensitivities for each stock, and returns time series of idiosyncratic stock returns. See section "Proposed Methodology" for more details.
- **visualize_idio_stock** accepts all the same parameters as above, as well as one stock ticker, and calls **idio_stock**, then visualizing the new time series of idiosyncratic returns of the ticker set by the user. On the same graph time series of observable returns for the chosen stock can also be plotted.

The second script is called stochastic_edge.py, and it contains a number of classes, corresponding to various stochastic processes, which share basic methods for their generation and calibration. Seven classes are present in the script:

- Zero-mean Algebraic Brownian motion
- Algebraic Brownian motion
- Geometric Brownian motion
- Vasicek model
- Heston model

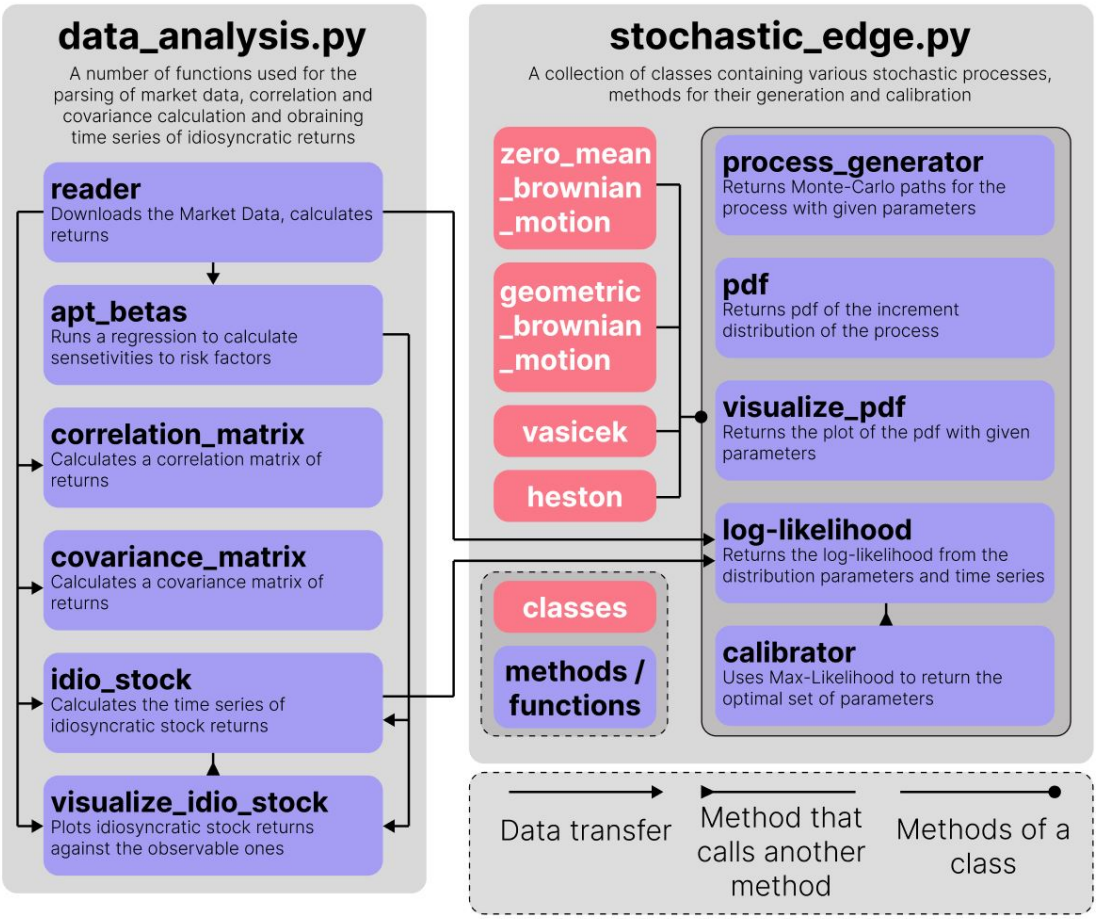
- Single-stock Merton process - a Jump Process discussed in the literature review section
- Multi-stock Merton process, which can model paths, while taking non-zero correlation between stocks into account

Each of the classes above contains a set of standard methods that share the same syntax across classes:

- **process_generator**, given a set of parameters specific to the process used, generates random paths of that process. Parameters of the process are passed from the initialization function of a class. User inputs the number of paths to generate, their length, whether the process must be visualized, and if so, should it be saved as a .jpg file, as well as whether the class should return paths' values or returns on each path.
- **pdf** given a set of parameters and x, return the value of the Probability Density Function of returns at x: $f(x, \theta)$.
- **visualize_pdf** returns a plot of the PDF (this method is limited to one-dimensional PDFs).
- **log_likelihood** takes as an input time series of returns and the parameters of the distribution and returns the sum of logs of PDFs at every point of historic returns: $L = \sum_{i=1}^{n} \log(f_i)$.
- **calibrator** is a method that performs the optimization on the log-likelihood function to find such set of distribution parameters that maximize it.

After finding the optimal parameters for the process, it is possible to pass these parameters into **process_generator** method to get a simulation of the calibrated process.

The overall structure of my scripts can be presented via a graph below:



Using that set of Python script, I calibrated and simulated all the risk factors, discussed above.

5 Modelling risk factors

In this chapter, I am going to discuss possible ways of modelling of the risk factors within my two-factor model. There are three reasons why each of the individual risk factors has to be fully modelled:

1. The necessity to price derivatives. For example, options' payoffs depend on observable prices, and, by extension, observable returns, not idiosyncratic ones. Therefore, it is important to model the overall dynamic of not only the idiosyncratic shocks of stocks, but also all risk factors, to be able to "reconstruct" the observable returns when performing Monte-Carlo pricing of the derivatives.

2. If the variance of each risk factor is known, it becomes possible to calculate the share of total risk in buying a stock that comes from each factor, and more broadly, understand the amounts by which different factors contribute to the overall risk of the stock.

3. Following the previous point, if the risk born from the factors can be quantified, it becomes possible to hedge it using market instruments. This allows trader to get exposure exclusively to the idiosyncratic performance of the stock, making their Mark-to-Market less volatile and removing macro-effects from their P&L.

Therefore, I becomes clear that each risk factor must be modelled in detail, which I decided to do using two stochastic models. This section will also cover the modelling of stock's idiosyncratic returns.

5.1 Risk-free rate

There are a number of models that can be used for estimation and subsequent simulation of the interest rate dynamics. If the problem requires the modelling of the entire yield curve, Heath–Jarrow–-Morton model, first proposed in a number papers [15], [16] and [17] can be used. However, for the purposes of my work, so-called instant (or continuous) rate, which can be viewed as overnight rate for US money market, should be used. For the purposes of data aggregation I am going to use ÎRX index, which tracks the yields of 13-week treasury bonds. Obviously, the usage of such proxy

introduces basis risk into my model, however, I am going to ignore it, as 13-week bond yields closely follow the SOFR due to their short duration.

The most popular model used for short-rate simulation is Vasicek, first proposed in [47]. The model is described by the Stochastic Differential Equation:

$$dr_{ft} = a(b - r_{ft})dt + \sigma_r dW_t \tag{45}$$

Where b is the long-term mean interest rate in the economy, a is the speed of mean reversion, and σ_r is the volatility of the interest rate. Before any further analysis, I decided to check whether the time series of ÎRX index is stationary, and if not, whether its returns series is. To determine the presence of weak stationarity in these time series I ran Augmented Dickey-Fuller test. The following results were obtained:

Time series	Test Statistic	P-value
IRX index	-0.3790	0.9135
IRX returns	-5.8067	0.0000

This indicates that the original time series of IRX index is not stationary (technically speaking, we cannot reject the hypothesis of non-stationarity), meanwhile, its returns are indeed stationary. This means that the Vasicek model is adequate for the purposes of modelling such time series, as weak stationarity implies constant expected value of the time series (which is achieved by the introduction of mean-reversion component into the model).

However, I decided to also test a simplified version of the Vasicek model, which does not exhibit mean-reverting property. Such model is equivalent to the symmetric Algebraic Brownian motion and can be represented by the following stochastic differential equation:

$$dr_{ft} = \sigma_r dW_t \tag{46}$$

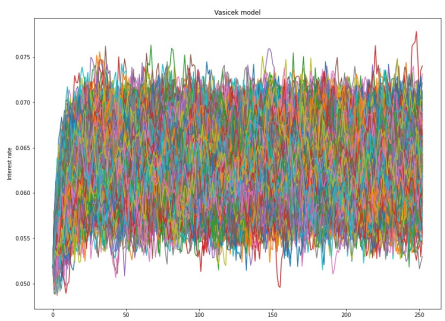
The models were calibrated by the Max-Likelihood method, which was previously described in details in this paper.

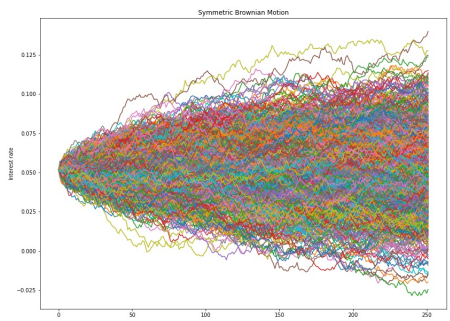
The following results were obtained for mean-reverting and non-mean-reverting models:

Parameter or metric	Vasicek model	Brownian Motion
a (speed of reversion)	0.1403	-
b (long-run mean)	0.0633	-
σ_r (volatility of the rate)	0.158335%	0.158465%
Log-Likelihood	318.5009	317.8868
Model training time	32.8s	7.9s

The results are somewhat unexpected: two different models not only converged to the same volatility of the interest rate (to the 5th decimal!), but they also yielded Log-likelihood, which is different by 0.2%. However, Vasicek model's training required more than 4 times the amount of compute time, compared to the basic Brownian motion. I find this case exemplary, as it shows, that often more sophisticated models can lead to near-zero gains in accuracy, while requiring significantly more resources to train and simulate random paths, and generally being more mathematically complex, which makes development and maintenance of new and existing models and code more difficult.

However, the two models yielded different patterns when simulating paths:





As my goal is predicting the change of interest rates over the relatively long time period (for instance, 1 year), I suggest using Zero-mean Brownian motion. The economic logic for it is simple: long-term rate in Vasicek model resembles current Fed rate, around which short rates fluctuate, and which stays constant during a short period. However, for one year simulation, it is reasonable to expect Fed rate to change, therefore, long-term rate is not constant, and it is better to use simple Brownian motion. It will serve as the part of the overall model responsible for simulating risk-free rate.

5.2 S&P500 returns and VIX

Importantly, in my setup both broad market returns and volatility comes from the same source: S&P500 index. Furthermore, by the methodology of VIX index [11], it is constructed from the same set of options on S&P500 in every period, ensuring that we stay on the same spots on the volatility surface. This means, that we can ignore the local volatility effect in favor of purely stochastic volatility model.

The most popular approach to building stochastic volatility is Heston model, which is defined as:

$$\begin{cases} dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^S \\ dv_t = \kappa(\theta - v_t)dt + \xi\sqrt{v_t} dW_t^v \end{cases} . \tag{47}$$

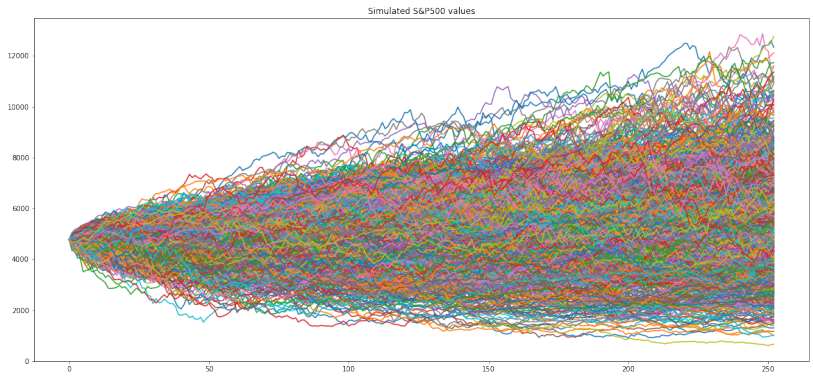
Where μ is the stock's drift term, θ is the long-term mean volatility, κ is the speed of mean reversion and ξ is the volatility of volatility. Note, that analogous to the Vasicek model, described above, Heston assumes the mean-reverting nature of volatility.

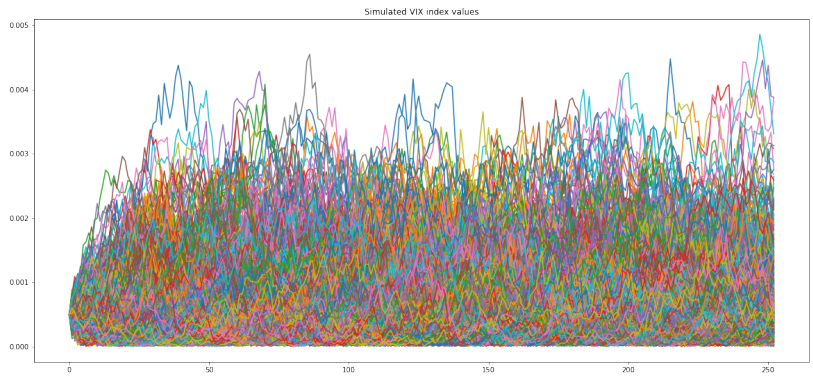
In order to simplify the optimization procedure, I am going to use the "naive" estimator for ρ - the correlation coefficient between returns of S&P500 and the returns of VIX. On the time series used ρ is estimated to be -0.7328. This finding was expected, as it is a well-known stylized fact that volatility is asymmetric: it increases by the larger amount when prices decrease. The mathematical foundation, as well as economic rationale behind this phenomenon were studied by Andrew A. Christie [5] and Geert Bekaert, Guojun Wu [27].

Using Max-Likelihood, I calibrated the model to the following results:

Parameter or metric	Vasicek model
μ (market drift in natural measure)	0.0025
κ (speed of mean reversion)	0.0321
θ (long-run level of mean reversion)	0.0007
ρ (correlation between S&P500 and VIX returns)	-0.7328
ξ (volatility of volatility)	0.0518

And the simulated paths of Heston model with these parameters are presented below:



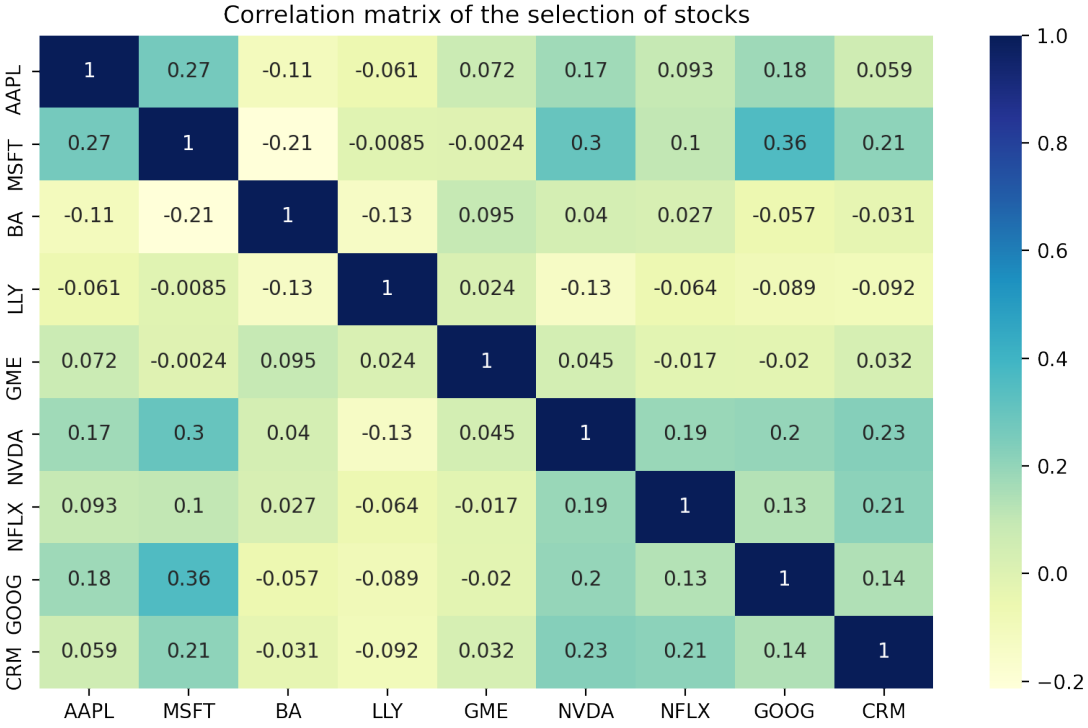


As I have sufficient approaches to modelling all risk factors within my APT-based framework, I can finally move to the study of idiosyncratic stock returns.

5.3 Stocks' idiosyncratic returns

In order to approach the task of modelling idiosyncratic stock returns, it is important to perform some simple statistical data analysis. As mentioned above, idiosyncratic returns exhibit a higher degree of kurtosis compared to the observable stock returns (in other words, they have heavier tails).

Next, it is expected that the idiosyncratic returns would be uncorrelated between each other, however, it is not the case. To study this effect, I constructed a correlation matrix:



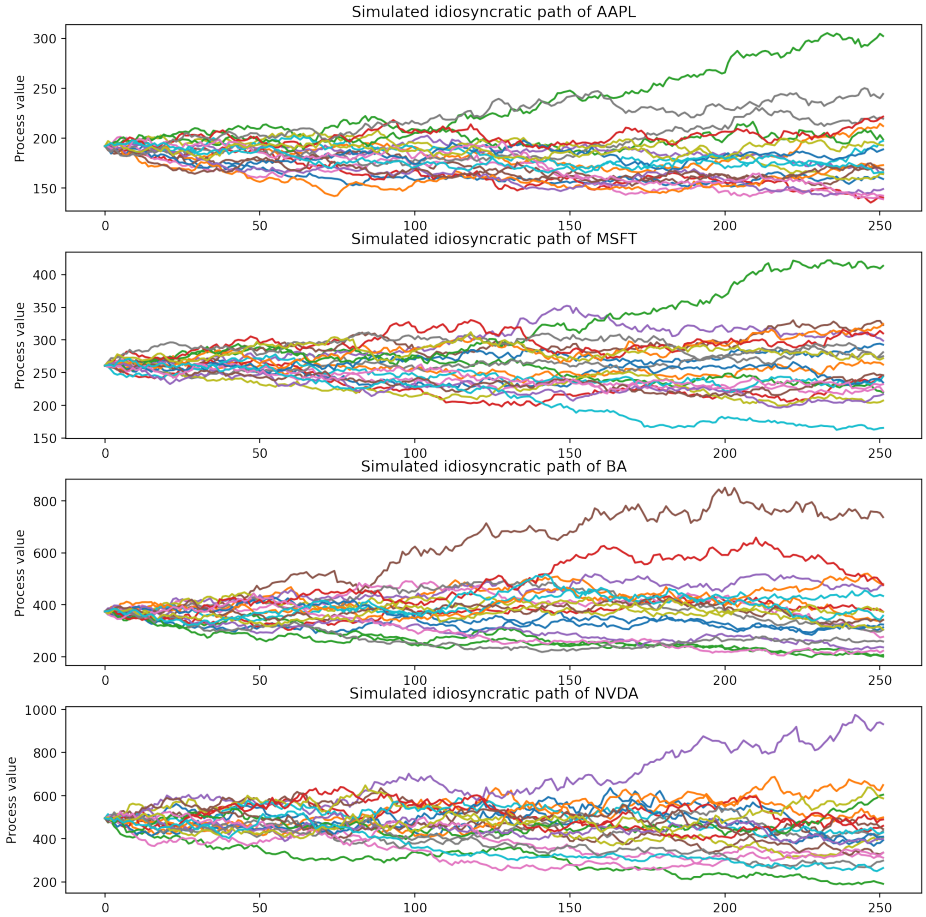
As can be seen, some stocks still exhibit statistically significant correlation of returns, even after removing reactions market-wide effects. By studying the correlation closer, I came to the conclusion, that correlated stocks usually fall into the same industry. For example, Microsoft (MSFT) and Apple (AAPL), Salesforce (CRM), Alphabet (GOOG) and NVidia (NVDA) all have pairwise correlation coefficient above 0.2, which can be considered statistically significant given a sample size of 752

(three years of daily trading data). Interestingly, Boeing (BA) and Microsoft have a statistically significant negative correlation between returns equal to -0.21. My only possible explanation of this phenomenon is that both Microsoft and Boeing are major government contractors, providing services for the American military, so they may be competing on some contracts, which yields significant negative correlation. In conclusion, even after removing market-wide effects stocks still exhibit statistically significant correlation, therefore, in order to properly calibrate the model it is necessary to take into account the interaction between stocks' returns.

In order to do this, I decided to calibrate a n-dimensional Multivariate correlated Merton Jump-diffusion process, which would take into account the rich cross-stock interactions, as well as leptokurtosis of the idiosyncratic stocks' returns. Generally, to generate the Multivariate Normal Distribution, two parameters must be provided: an vector of mean values and a n-dimensional covariance matrix - this is analogous to the Univariate Normal Distribution, which takes mean and variance as the input. Therefore, the process of calibration of the model involves finding the mean vector and covariance matrix that maximize log-likelihood, when the model is inputted the historic returns data.

However, in order to reduce computational complexity of the model, I am going to fix the correlations between stock returns and reduce the problem to finding optimal vectors of means, covariance matrices and the probability of a jump (λ) for the historic data. Even with that, this optimization problem is significantly computationally complex: the model trained on a selection of just 4 stocks finished calibration in more than an hour on my PC.

The visualization of the paths generated with the MLE-optimized parameters, is presented below:



For this dataset, the results of the calibration showed insignificant improvement in the in-sample back-testing, compared to simpler and less computationally expensive models:

Model	Log-Likelihood	Model calibration time
Correlated Multivariate Merton	8297	61 minutes
Uncorrelated Multivariate Merton	7803	60 minutes
Multivariate Normal	7589	16 minutes

Therefore, a trade-off between the accuracy of the model and calibration time arises. This, alongside other computational concerns, will be discussed in the following section.

6 Relative computational efficiency of the Max-Likelihood estimator

The model calibration via Max-Likelihood estimation is highly computationally expensive. It suffers from the curse of dimensionality, in other words, the time required to train a model rapidly increases with each new parameter. Calibration of the largest model - multivariate Merton jump-diffusion process, which, when optimized, has $4n + 1$ parameters to calibrate, and when trained with the "naive" approach of optimizing each entry in the covariance matrix - $2n^2 + 2n + 1$ parameters (n is the number of stocks the model is trained on).

It took me over an hour to train the model on the dataset of 3 years (752 trading days) for only 4 stocks. Therefore, it is appears unrealistic to train the any "large" model on the full set of parameters.

To avoid the curse of dimensionality, I found two alternative approaches to decrease computational complexity of the optimization. The first is to either set by hand or pre-calculate some of the parameters of the model. For example, set $\mu = 0$ (which is reasonable, as the APT-based algorithm removes risk-free rate from the drift in risk-neutral setting, therefore, making drift term infinitesimally small) or use correlation coefficients ρ, calculated directly from the time series. However, it is possible to take this approach one step further: instead of using max-likelihood, is it possible to use Method of Moments to estimate parameters? The next subsection discusses this problem in details.

6.1 Model Calibration using Method of Moments

To begin my analysis of the effectiveness of the Method of Moments in comparison to Max-Likelihood estimation, I decided to calibrate two parameters (μ, σ) of the Geometric Brownian motion for the idiosyncratic returns of a number of stocks. The results are presented below.

Stock	MLE Parameters	MoM parameters	MLE Log-likelihood	MoM Log-Likelihood
AAPL	(0, 0.0104)	(0, 0.0104)	2370.0899	2370.0896
MSFT	(0, 0.0108)	(0, 0.0108)	2336.1454	2336.1451
BA	(0, 0.0192)	(0, 0.0192)	1904.2856	1904.2853
GOOG	(0, 0.0133)	(0, 0.0133)	2179.3179	2179.3176
NVDA	(0, 0.0228)	(0, 0.0228)	1775.7169	1775.7165
GME	(0, 0.0955)	(0, 0.0955)	699.6639	699.6635

As can be seen in the table above, for all stocks examined, the difference in the Log-likelihood between MLE and MoM-estimated parameters is within the 4th decimal. The obtained parameters of the process are identical to the 4th decimal, although they obviously differ, as Log-likelihoods resulting from the use of two methods discussed are not equal. However, the calibration time for MoM is near instant, meanwhile MLE takes up to 10 seconds to calibrate two parameters of the model on the 3 years of data.

Moving further, I decided to examine the differences the method of model calibration makes on the Log-Likelihood of Merton jump-diffusion model.

6.1.1 Method of Moments for Calibration of One-dimensional Merton Process

Importantly, it is impossible to apply the same method to the jump diffusion processes, as they contain two sources of randomness. If the same set of parameters is passed into both distributions used, in case of Merton model, it simply becomes a regular Normal distribution with given parameters, which eliminates the purpose of the Jump-diffusion process, as such construction does not account for stylized facts of the stock returns (including heavy tails and left skew).

I propose a simple, yet effective numerical method for calibrating jump-diffusion model on the historical time series of returns. For simplicity I am going to call it Split-MoM. It can be done in four steps:

1. Calculate the mean and standard deviation of the original time series of returns.
2. Split the time series: all the returns that lie within 2σ of the mean fall into one sample ("non-jump"), the other returns fall into another one ("jump").
3. Finally, calculate new means and standard deviations for these two samples. Use them as the parameters of the model.
4. The probability of a jump is estimated as the number of elements in the "jump" sample, divided by the length of the original time series of returns.

Such simple approach yield relatively good results: the average difference between the Log-Likelihood of the Max-Likelihood estimated model (which took on average more than 2 minutes to run) and Split-MoM approach is 0.52%. The table below shows the details:

Stock	MLE Log-likelihood	Split-MoM Log-Likelihood	Difference in Log-Likelihood
AAPL	2414.7217	2405.1927	0.4%
MSFT	2403.5536	2400.1758	0.14%
BA	1944.0062	1934.1305	0.51%
GOOG	2274.7307	2266.0934	0.38%
NVDA	1825.3581	1822.0911	0.18%
GME	1087.0146	1070.5373	1.54%

Therefore, it is possible to conclude that the proposed Split-MoM algorithm is an effective alternative to the Max-Likelihood estimation when calibrating One-dimensional Merton model.

6.1.2 Method of Moments for Calibration of Multi-dimensional Merton Process

The similar technique could be used to calibrate the Multi-variable Merton process. The only difference is that it is required to generate a covariance matrix to get a PDF of the multivariate distribution. I suggest the following procedure:

1. Calculate a correlation matrix for returns of between stocks.
2. Split each individual time series of returns into two samples: inside 2σ and outside of them, analogous to the Split-MoM algorithm, described for the One-dimensional Merton model.
3. Calculate means and standard deviations for both samples for each stock.
4. Get covariance matrices for "Non-jump" (inside 2σ) and "Jump" (outside 2σ) samples by multiplying each entry in the array of correlation matrix by the standard deviations of two samples of stocks that are covariated.
5. Use the obtained covariance matrices alongside arrays of means for two samples to generate the PDF or paths of the distribution.

This method enjoys high computational efficiency, however, the results of the calibration are inferior, compared to both Max-Likelihood estimation and even correlated Geometric Brownian Motion (which does not capture heavy tails of the empirical distribution). In my future works, I may propose different approach of using Method of Moments for the calibration of Multivariate Merton Jump-diffusion process, however, now I am going to suggest using either full Max-Likelihood-calibrated model or its Normal analogue, which has a much lower computational complexity.

7 Conclusion

With this paper, I achieved a number of key goals I had set for myself. Firstly, I developed a APT-style model that allowed me to remove market wide effects from the observable stock returns, which was necessary for the accurate modelling of the idiosyncratic shocks, such as dividend gaps. Secondly, I packaged all the functions I have written for the data analysis done for this paper into a library, that can be easily used for a number of other applications. Thirdly, I created a library **stochastic_edge.py**, which contained a number of popular stochastic processes as classes, with unified method system, that can be flexibly used for the model calibration and path generation. Finally, I tested the effectiveness of the Max-Likelihood estimation against Method of Moments estimation. The conclusion I came to was that Method of Moments offers a less accuracy (for most models, the difference is less than 1%), when in-sample back-test is performed, however, it is much less computationally expensive - the difference may be by the factor of thousands.

Jump-diffusion processes proved to be capable of more accurately reflecting the stylized facts of the equity markets, in particular, the heavy-tailed and left-skewed distribution of returns. I have successfully developed algorithms of Max-Likelihood and Method of Moments for their calibration, which can be used in the real-world setting.

All the scripts used in this paper will be attached alongside this file.

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